Nonlinear Generalized Monotonicity and Error Bounds for Proximal Point Convergence at Multivalued Graph Junctions

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Abstract

Quadratic generalized monotonicity conditions control proximal mappings by Lipschitz estimates. Multivalued graph junctions can instead produce a Hölder, non-Lipschitz proximal response. We introduce an all-pairs nonlinear RL graph inequality that accommodates this behavior and combine it with an independent generalized metric-subregularity gauge. Under standard local range coverage, RL makes the represented proximal mapping single-valued and gives both a coupled energy estimate and the sharp step bound $s \le (d+Ld^\gamma)/2$, where d is the distance to the zero set. The error bound then yields the scalar recursion $d^+ \le \psi((d+Ld^\gamma)/(2\lambda))$. Strict local contraction of this map gives distance contraction, an explicit localization radius, finite length, and convergence of the whole proximal orbit to one point of a possibly nonisolated zero set. The same proof also gives a selection-uniform extension for set-valued proximal relations. For residual growth $\operatorname{dist}(0, F(y)) \ge m \operatorname{dist}(y, S)^a$, with $L > 0$ and $0 < \gamma < 1$, compatibility holds when $a < \gamma$; at the critical order $a = \gamma$, the direct threshold is $\lambda m > L/2$. A two-valued square-root operator attains the optimal exponent $1/2$ together with a quadratic error bound. Its proximal orbit has exact distance factor $1/4$, whereas every finite exponent-one RL, Lipschitz J-side, and local hypomonotonicity certificate fails on the same full neighborhood. This gives an explicit multivalued regime in which nonlinear RL supplies the missing operator geometry and the error bound converts it into PPA convergence.

Keywords. proximal point algorithm; generalized monotonicity; metric subregularity; set-valued operator; Hölder proximal mapping.

1 Introduction

Let $F : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ and $\lambda > 0$. The proximal point algorithm (PPA) seeks a zero of F through the iteration

$$x^{k+1} \in J_{\lambda F}(x^k), \qquad J_{\lambda F} := (I + \lambda F)^{-1}. \tag{1}$$

For monotone operators, the inner-product inequality on the graph gives firm nonexpansiveness of $J_{\lambda F}$ and the associated decrease estimate [11]. Generalized-monotonicity approaches extend this geometry beyond monotone operators and combine it with residual error bounds to prove local convergence [7, 8, 2]. We develop this approach for a nonlinear graph inequality and a general metric-subregularity gauge.

The obstruction addressed here occurs when several graph branches meet. A multivalued operator can have a well-defined proximal mapping whose response to an input perturbation of size t is of order $t^{1/2}$. The mapping is continuous but not locally Lipschitz, so no finite quadratic firm-type certificate holds on a full neighborhood of the junction. A useful

operator condition must control the complete graph across branches at the Hölder scale and still produce a step estimate that can be paired with an error bound. Nonlinear RL is designed for exactly this role.

For graph points $(u, u^*), (v, v^*)$ of F, write $a = u - v$ and $b = u^* - v^*$. We impose the nonlinear *RL condition*

$$\lambda\langle a, b\rangle \geq \frac{1}{4}\left(\|a + \lambda b\|^2 - L^2\|a + \lambda b\|^{2\gamma}\right), \qquad L \geq 0, \quad 0 < \gamma \leq 1. \tag{2}$$

The condition is required for every pair in the specified graph localization, including pairs from different branches. At $\gamma = 1$ and $L = 1$ it reduces to monotonicity. The general exponent-one case is a quadratic generalized-monotonicity condition; $\gamma < 1$ allows a non-quadratic lower bound on the same graph inner product.

The form used in the PPA analysis is an equivalent J-side inequality. For $x^+ \in J_{\lambda F}(x)$ and $y^+ \in J_{\lambda F}(y)$ whose graph points belong to the localization, (2) gives

$$\|x^+ - y^+\|^2 + \|(x - x^+) - (y - y^+)\|^2 \leq \frac{\|x - y\|^2 + L^2\|x - y\|^{2\gamma}}{2}.$$

This J-side inequality controls both the proximal output and the proximal residual. Taking $x = y$ proves uniqueness of every represented proximal fiber. We combine it with the standard local range-coverage condition used in nonmonotone PPA theory, which supplies an output for every input in the working neighborhood. The main theorem therefore starts from one assumption block containing coverage, RL geometry, the error bound, and their scalar compatibility. A subsequent extension shows that the same argument remains uniform over a set-valued proximal relation whenever the one-step estimates hold for every allowed choice.

The second ingredient is generalized metric subregularity:

$$\operatorname{dist}(y, S) \leq \psi\big(\operatorname{dist}(0, F(y))\big), \qquad \varnothing \neq S \subset F^{-1}(0), \tag{3}$$

where S is closed and ψ is nondecreasing, with $\psi(0) = 0$ and $\psi(t) \to 0$ as $t \downarrow 0$. This assumption recovers distance to S from the residual and therefore permits nonisolated solutions. It is verified independently of the graph geometry. Slope criteria, generalized differentiation, and higher-order subregularity provide sources of such estimates [3, 9, 5]. We call the coupling of RL geometry with this independent regularity input the RL–error-bound (RLEB) framework.

The proof combines these two ingredients directly. For a proximal transition, put

$$d = \operatorname{dist}(x, S), \qquad d^+ = \operatorname{dist}(x^+, S), \qquad s = \|x - x^+\|.$$

Comparison with a nearest solution yields

$$(d^+)^2 + s^2 \leq \frac{d^2 + L^2 d^{2\gamma}}{2}, \qquad s \leq \frac{d + L d^\gamma}{2}. \tag{4}$$

The factor $1/2$ in the step estimate follows by completing the square in the J-side inequality. Since $(x - x^+)/\lambda \in F(x^+)$, the error bound gives

$$d^+ \leq \psi\left(\frac{d + L d^\gamma}{2\lambda}\right). \tag{5}$$

If the right-hand side is at most κd near zero for some $\kappa < 1$, the distance decreases geometrically. Substitution into the step estimate gives a summable sequence of step lengths.

Its sum controls the region visited by the iterates, allows repeated use of local proximal existence, and proves convergence to one solution.

For the power error bound arising from $\operatorname{dist}(0, F(y)) \geq m \operatorname{dist}(y, S)^a$, the direct recursion has leading order $d^{\gamma/a}$ when $L > 0$ and $0 < \gamma < 1$. Thus $a < \gamma$ gives local compatibility, while the critical case $a = \gamma$ requires $\lambda m > L/2$. These are criteria for the displayed scalar recursion. A complementary use of the coupled energy estimate in (4) gives additional parameter regimes, including the standard metric-subregularity factor at the monotone endpoint.

The principal results are the following.

1. A graph-level theorem combines standard local range coverage, nonlinear all-pairs RL, and a general error-bound gauge. It gives distance contraction, finite length, an explicit localization budget, and convergence of the whole orbit to a point of a possibly nonisolated zero set. A selection-uniform corollary covers multivalued proximal relations under the corresponding transition estimates.

2. The step estimate retains the coefficient $1/2$ and gives an explicit compatibility test for the two independent hypotheses. For power residual growth, the test determines the critical order and the threshold $\lambda m > L/2$. The energy estimate provides a complementary test and calibrates the linear endpoint against existing PPA rates.

3. A two-valued square-root operator has optimal RL exponent $1/2$ and a quadratic residual error bound. The theorem gives a positive-radius convergence guarantee whose distance factor approaches the exact factor $1/4$. The complete orbit is explicit, and its pointwise asymptotic factor is $1/2$. Direct graph and J-side comparisons identify the quadratic certificates that fail on the same neighborhood.

When a formula for $J_{\lambda F}$ is unavailable, we provide a derivative-based route for checking the graph hypotheses. Classical prox-regularity and coderivative criteria already supply useful generalized-monotonicity information [10, 1]. For a complete two-sided graph junction with one degenerate normal direction, the signed-Schur test developed here uses tangential invertibility and signed normal growth to verify local coverage, cross-branch uniqueness, and explicit RL parameters. The matching error bound is supplied by the regularity theory appropriate to the operator.

Section 2 develops the preliminary estimates and the convergence theorem. Section 3 gives the square-root example, and Section 4 identifies its relation to quadratic generalized-monotonicity theories. Section 5 presents the signed-Schur test; its full hypotheses and proof are collected in the appendix.

2 RL generalized monotonicity and local PPA convergence

Throughout, $X = \mathbb{R}^n$ has its Euclidean inner product and norm, $F : X \rightrightarrows X$, and $\lambda > 0$. We write

$$J_{\lambda F}(x) := \{u \in X : x - u \in \lambda F(u)\}, \qquad r_F(y) := d(0, F(y)).$$

The solution set under consideration is a nonempty closed set $S \subset F^{-1}(0)$. Its metric projection $P_S(x)$ is nonempty, although it need not be single-valued. For any set $A \subset X$, we use $d(x, A) := \inf_{a \in A} \|x - a\|$ and $d(x, \varnothing) := +\infty$.

2.1 RL and its J-side form

Definition 2.1 (Hölder RL generalized monotonicity)**.** Let $\mathcal{G} \subset \operatorname{gph} F$, $R > 0$, $L \geq 0$, and $0 < \gamma \leq 1$. We say that $\mathcal{G}$ satisfies $\mathrm{RL}(\lambda, \gamma, L)$ at scale R if, for every $(u, u^*), (v, v^*) \in \mathcal{G}$, with

$$a := u - v, \qquad b := u^* - v^*, \qquad \|a + \lambda b\| \leq R,$$

one has

$$\lambda \langle a, b \rangle \geq \frac{1}{4} \left(\|a + \lambda b\|^2 - L^2 \|a + \lambda b\|^{2\gamma} \right). \tag{6}$$

Equivalently,

$$\|a - \lambda b\| \leq L \|a + \lambda b\|^{\gamma}. \tag{7}$$

When λ is fixed, we abbreviate $\mathrm{RL}(\lambda, \gamma, L)$ to $\mathrm{RL}(\gamma, L)$.

The quantifier includes pairs on different branches of F. At $\gamma = 1$ and $L = 1$, (6) is the monotonicity inequality for the tested graph pairs.

For a local graph set $\mathcal{G}$, define the associated proximal relation by

$$J_{\mathcal{G}}(x) := \{u : (u, (x - u)/\lambda) \in \mathcal{G}\}.$$

It agrees with $J_{\lambda F}$ on an input region whenever all proximal graph points over that region belong to $\mathcal{G}$.

Lemma 2.2 (Resolvent form of RL)**.** *The condition in Definition 2.1 is equivalent to*

$$\|u - v\|^2 + \|(x - u) - (y - v)\|^2 \leq \frac{1}{2} \left(\|x - y\|^2 + L^2 \|x - y\|^{2\gamma} \right) \tag{8}$$

for every $u \in J_{\mathcal{G}}(x)$ and $v \in J_{\mathcal{G}}(y)$ with $\|x - y\| \leq R$. In particular, $J_{\mathcal{G}}$ is single-valued on its domain. Assumption 2.4 below fixes that domain through the standard local range-coverage condition.

Proof. For the corresponding graph points, put $a = u - v$ and $\lambda b = (x - u) - (y - v)$. Then $a + \lambda b = x - y$ and

$$\|a\|^2 + \lambda^2 \|b\|^2 = \|a + \lambda b\|^2 - 2\lambda \langle a, b \rangle.$$

Substitution proves the equivalence. Setting $x = y$ in (8) gives $u = v$. □

Definition 2.3 (Generalized metric subregularity)**.** Let $\bar{t} > 0$ and let $\psi : [0, \bar{t}] \to [0, \infty)$ be nondecreasing, with $\psi(0) = 0$ and $\psi(t) \to 0$ as $t \downarrow 0$. The mapping F satisfies the ψ-error bound relative to S on V if

$$d(y, S) \leq \psi\big(r_F(y)\big) \qquad (y \in V,\ r_F(y) \leq \bar{t}). \tag{9}$$

2.2 The local RLEB assumptions

For $(u, u^*) \in \mathcal{G}$, let

$$M_+(u, u^*) := u + \lambda u^*, \qquad M_+(\mathcal{G}) := \{M_+(u, u^*) : (u, u^*) \in \mathcal{G}\}.$$

Thus $M_+(\mathcal{G})$ is exactly the set of proximal inputs represented by the graph block. The following assumption places the full local PPA setting in one statement.

Assumption 2.4 (Local RLEB data)**.** Fix an open set $U \subset X$, numbers

$$R > 0, \quad \bar{t} > 0, \quad L \geq 0, \quad 0 < \gamma \leq 1, \quad 0 < \kappa < 1,$$

and a gauge ψ as in Definition 2.3. Put $U_R := \{x \in U : d(x, S) \leq R\}$. Assume:

(A1) *Local range coverage.* $U_R \subset M_+(\mathcal{G})$, and for every $x \in U_R$ there is $p \in P_S(x)$ with $(p, 0) \in \mathcal{G}$.

(A2) *Nonlinear graph geometry.* The graph block $\mathcal{G}$ satisfies $\mathrm{RL}(\lambda, \gamma, L)$ at scale R.

(A3) *Output error bound.* The estimate (9) holds at every $y \in J_{\mathcal{G}}(U_R)$ with $r_F(y) \le \bar{t}$, and

$$\frac{R + LR^\gamma}{2\lambda} \le \bar{t}. \tag{10}$$

(A4) *RLEB compatibility.*

$$\psi\left(\frac{r + Lr^\gamma}{2\lambda}\right) \le \kappa r \qquad (0 < r \le R). \tag{11}$$

Condition (A1) is the standard local range-coverage requirement used in nonmonotone PPA theory: compare, for example, the inclusion $U' \subset (I + \lambda \bar{F})(U)$ in [7, Assumption 2(d)(ii)]. Assumption (A1) adopts this established well-posedness interface. Assumption (A2) makes each represented fiber single-valued by Lemma 2.2; hence (A1) defines a unique localized proximal mapping $J_{\mathcal{G}}$ on U_R. The nonlinear extension is carried by (A2)–(A4).

Lemma 2.5 (The RLEB one-step estimates)**.** *Assume* (A1)–(A3). *Let $x \in U_R$, set $x^+ := J_{\mathcal{G}}(x)$, and define*

$$d := d(x, S), \qquad d^+ := d(x^+, S), \qquad s := \|x - x^+\|.$$

Then

$$(d^+)^2 + s^2 \le \frac{d^2 + L^2 d^{2\gamma}}{2}, \tag{12}$$

$$s \le \frac{d + Ld^\gamma}{2}, \tag{13}$$

$$d^+ \le \psi(s/\lambda), \tag{14}$$

$$d^+ \le \psi\left(\frac{d + Ld^\gamma}{2\lambda}\right). \tag{15}$$

Proof. Choose $p \in P_S(x)$ as in (A1). Apply RL to the proximal graph point $(x^+, (x - x^+)/\lambda)$ and $(p, 0)$. The resulting J-side inequality gives

$$\|x^+ - p\|^2 + s^2 \le \frac{d^2 + L^2 d^{2\gamma}}{2},$$

which proves (12). With $z = x - p$ and $v = x - x^+$, the same inequality and $\langle z, v \rangle \le ds$ yield

$$d^2 - 2ds + 2s^2 \le \frac{d^2 + L^2 d^{2\gamma}}{2}.$$

Completing the square gives

$$(2s - d)^2 \le L^2 d^{2\gamma}, \qquad s \le \frac{d + Ld^\gamma}{2}. \tag{16}$$

Finally, $(x - x^+)/\lambda \in F(x^+)$ implies $r_F(x^+) \le s/\lambda$. Assumption (A3), monotonicity of ψ, and (13) prove the last two estimates. □

2.3 The local convergence theorem

Theorem 2.6 (Local convergence under nonlinear RL and a generalized error bound)**.** *Let Assumption 2.4 hold. Let $x^0 \in U$, put $d_0 = d(x^0, S) \le R$, and assume*

$$\mathcal{L}(d_0) := \frac{1}{2}\left(\frac{d_0}{1-\kappa} + \frac{L d_0^\gamma}{1-\kappa^\gamma}\right) < \operatorname{dist}(x^0, X \setminus U). \tag{17}$$

Then the localized PPA orbit

$$x^{k+1} = J_{\mathcal{G}}(x^k) \tag{18}$$

can be continued indefinitely, remains in $U \cap \{x : d(x,S) \le R\}$, has finite length, and converges to a point $x^\infty \in S$. More precisely,

$$d(x^k, S) \le \kappa^k d_0, \tag{19}$$

$$\|x^{k+1} - x^k\| \le \frac{1}{2}\left(\kappa^k d_0 + L\kappa^{\gamma k} d_0^\gamma\right), \tag{20}$$

$$\|x^\infty - x^k\| \le \frac{1}{2}\left(\frac{\kappa^k d_0}{1-\kappa} + \frac{L\kappa^{\gamma k} d_0^\gamma}{1-\kappa^\gamma}\right). \tag{21}$$

Proof. Suppose that $x^k \in U$ and $d_k := d(x^k, S) \le R$. Assumption (A1) supplies the next point, and (A2) makes it unique. Lemma 2.5 and (A4) give

$$d_{k+1} \le \psi\left(\frac{d_k + L d_k^\gamma}{2\lambda}\right) \le \kappa d_k.$$

This inequality also holds when $d_k = 0$, since (13) then forces $x^{k+1} = x^k$. Consequently, along every finite orbit, $d_k \le \kappa^k d_0$. The step estimate gives

$$s_k := \|x^{k+1} - x^k\| \le \frac{1}{2}\left(\kappa^k d_0 + L\kappa^{\gamma k} d_0^\gamma\right).$$

For each such orbit, every partial sum of the step lengths is at most $\mathcal{L}(d_0)$. By (17), its next point therefore remains in U. This closes the induction, establishes indefinite continuation, and proves (19)–(20).

The two geometric series are summable, so the orbit has finite length and is Cauchy. Summing from index k gives (21). Its limit belongs to S because S is closed and $d(x^k, S) \to 0$. □

Corollary 2.7 (Selection-uniform extension)**.** *Fix the parameters and sets appearing in Assumption 2.4, and let $\mathcal{J}(x) \subset J_{\lambda F}(x)$ be an allowed proximal relation. In place of (A1)–(A3), suppose that the following transition-level conditions hold on U_R:*

1. *$\mathcal{J}(x)$ is nonempty for every $x \in U_R$;*
2. *for every $x^+ \in \mathcal{J}(x)$ there is $p \in P_S(x)$ such that*

$$\langle x^+ - p, x - x^+ \rangle \ge \frac{1}{4}\left(d(x,S)^2 - L^2 d(x,S)^{2\gamma}\right); \tag{22}$$

3. *the error bound (9) holds at every allowed output, and (10) holds.*

If (A4) and (17) hold, then every sequence $x^{k+1} \in \mathcal{J}(x^k)$ satisfies all conclusions of Theorem 2.6. The limit may depend on the choices.

Proof. The proof of Lemma 2.5 uses graph-level RL only to obtain (22). Under the displayed transition-level conditions, the same energy, step, and output estimates hold uniformly for every allowed choice. The induction and summation in Theorem 2.6 therefore apply verbatim. □

For local use, the compatibility requirement is equivalently expressed by

$$\limsup_{r\downarrow 0} \frac{\psi\big((r+Lr^{\gamma})/(2\lambda)\big)}{r} < 1. \tag{23}$$

Indeed, this condition provides $\kappa \in (0,1)$ and a sufficiently small radius for (11). The gauge domain restriction in (A3) also holds after decreasing the radius.

Corollary 2.8 (Residual growth and the critical order)**.** *Assume (A1)–(A2), with* $0 < \gamma < 1$ *and* $L > 0$*. Suppose that, on the relevant output region,*

$$r_F(y) \geq m\, d(y,S)^a, \qquad m > 0, \qquad a > 0. \tag{24}$$

Then $\psi(t) = (t/m)^{1/a}$ *is an admissible error-bound gauge. Its direct compatibility condition has the following cases:*

1. *if* $a < \gamma$, (23) *holds;*
2. *if* $a = \gamma$, *it holds exactly when*
$$\lambda m > \frac{L}{2}; \tag{25}$$
3. *if* $a > \gamma$, *it fails.*

In the first two compatible cases, the local convergence theorem applies after choosing the radius and initial point as specified there. Failure of this scalar certificate alone makes no assertion about divergence.

Proof. The quotient in (23) becomes

$$\frac{1}{r}\left(\frac{r+Lr^{\gamma}}{2\lambda m}\right)^{1/a}.$$

Its limit is zero for $a < \gamma$, equals $(L/(2\lambda m))^{1/\gamma}$ for $a = \gamma$, and is infinite for $a > \gamma$. □

2.4 A complementary energy estimate

The coupled inequality (12) provides another convergence criterion. For this subsection, assume that ψ is continuous and strictly increasing on $[0,\bar{t}]$ and that $R \leq \psi(\bar{t})$. Define

$$\begin{gathered} h(r) := \lambda^2[\psi^{-1}(r)]^2, \qquad V(r) := r^2 + h(r) \quad (0 \leq r \leq \psi(\bar{t})), \\ A(r) := \frac{r^2 + L^2 r^{2\gamma}}{2}. \end{gathered} \tag{26}$$

Proposition 2.9 (Energy criterion)**.** *Assume* (A1)–(A3) *and the conditions of this subsection. Suppose that, for some* $q \in (0,1)$*,*

$$q_E(R) := \sup_{0<r\leq R} \frac{A(r)}{V(r)} \leq q. \tag{27}$$

If $x^0 \in U$, $d_0 \le R$, and

$$\frac{\sqrt{qV(d_0)}}{1-\sqrt{q}} < \operatorname{dist}(x^0, X \setminus U), \tag{28}$$

then the localized proximal orbit can be continued indefinitely, stays in the stated input region, has finite length, and converges to a point $x^\infty \in S$. It satisfies

$$V(d_k) \le q^k V(d_0), \qquad d_k \le q^{k/2}\sqrt{V(d_0)}, \tag{29}$$

$$\|x^{k+1} - x^k\| \le q^{(k+1)/2}\sqrt{V(d_0)}, \tag{30}$$

$$\|x^\infty - x^k\| \le \frac{q^{(k+1)/2}\sqrt{V(d_0)}}{1-\sqrt{q}}. \tag{31}$$

Proof. For a transition from $d_k \le R$, (A3) and the step bound give $s_k/\lambda \le \bar{t}$. Hence $d_{k+1} \le \psi(s_k/\lambda) \le \psi(\bar{t})$, so $V(d_{k+1})$ is defined. Applying ψ^{-1} to the error bound yields $h(d_{k+1}) \le s_k^2$. Therefore

$$V(d_{k+1}) \le d_{k+1}^2 + s_k^2 \le A(d_k) \le qV(d_k).$$

Since V is strictly increasing, this also gives $d_{k+1} \le R$. Moreover, $s_k^2 \le A(d_k) \le qV(d_k)$. Iterating these inequalities gives (29)–(30). The budget (28) keeps every next iterate in U. Summation proves finite length and (31); closedness of S identifies the limit. □

For a general gauge, the energy criterion gives Q-linear decay of $V(d_k)$ and an R-linear upper bound for d_k. A distance Q-linear factor requires additional information about V, as in the linear specialization below. For (24), $h(r) = \lambda^2 m^2 r^{2a}$. At $a = \gamma < 1$,

$$\lim_{R\downarrow 0} q_E(R) = \frac{L^2}{2\lambda^2 m^2}, \tag{32}$$

and the energy condition holds precisely when

$$\lambda m > \frac{L}{\sqrt{2}}. \tag{33}$$

The direct condition (25) thus includes the additional range $L/2 < \lambda m \le L/\sqrt{2}$ at the nonlinear critical order.

Remark 2.10 (Scope of the scalar conditions). The conditions above test the displayed scalar bounds. They are sufficient conditions for convergence and do not assert necessity for an individual proximal orbit. In particular, (33) is not the best sufficient threshold under the full one-step geometry, because the step estimate also gives (25).

2.5 Power and linear specializations

Corollary 2.11 (Power gauges)**.** *Let $\psi(t) = Kt^\theta$, with $K > 0$ and $\theta > 0$. The direct compatibility map is*

$$\Phi(r) = \frac{K}{(2\lambda)^\theta}\big(r + Lr^\gamma\big)^\theta. \tag{34}$$

Under (A1)–(A3), the condition $\sup_{0<r\le R} \Phi(r)/r < 1$ supplies the compatibility assumption of Theorem 2.6. The energy function uses $h(r) = \lambda^2 K^{-2/\theta} r^{2/\theta}$, and its criterion becomes

$$\sup_{0<r\le R} \frac{r^2 + L^2 r^{2\gamma}}{2\big(r^2 + \lambda^2 K^{-2/\theta} r^{2/\theta}\big)} < 1. \tag{35}$$

Each criterion is used with its respective localization budget.

Corollary 2.12 (The linear slice)**.** *Let $\gamma = 1$ and suppose that (A1)–(A2) and the linear output error bound*

$$d(y, S) \le \rho\, r_F(y), \qquad \rho > 0. \tag{36}$$

The direct step estimate gives the distance factor

$$\kappa_H = \frac{(1+L)\rho}{2\lambda}, \tag{37}$$

whereas the coupled energy estimate gives

$$\kappa_E = \sqrt{\frac{1+L^2}{2}}\,\frac{\rho}{\sqrt{\rho^2+\lambda^2}}. \tag{38}$$

Either factor below one yields local finite-length convergence under the corresponding domain conditions and localization budget. Their strict-contraction conditions are, respectively,

$$(1+L)\rho < 2\lambda, \qquad L^2 < 1 + \frac{2\lambda^2}{\rho^2}.$$

At $L = 1$, the second factor is $\rho/\sqrt{\rho^2+\lambda^2}$, recovering the metric-subregularity PPA factor of Leventhal [4].

Proof. The first estimate follows from $d^+ \le (\rho/\lambda)s \le \kappa_H d$. For the second, $s \ge (\lambda/\rho)d^+$ and (12) give

$$\left(1 + \frac{\lambda^2}{\rho^2}\right)(d^+)^2 \le (d^+)^2 + s^2 \le \frac{1+L^2}{2}d^2.$$

Thus $d^+ \le \kappa_E d$. With either contracting factor, the step estimate and the summation argument of Theorem 2.6 give the stated orbit conclusions. □

3 A multivalued example beyond quadratic J-side estimates

The following two-branch operator separates the distance to the solution set from the displacement of a proximal step. Its proximal mapping contracts the normal distance by a factor of $1/4$, while its tangential displacement has square-root size. We verify the complete graph condition and the error bound separately, apply the convergence theorem, and then calculate the orbit to check the resulting estimate.

3.1 The operator and its complete proximal mapping

Theorem 3.1 (A two-branch square-root seam)**.** *Define $F : \mathbb{R}^2 \rightrightarrows \mathbb{R}^2$ by*

$$F(\xi, y) = \begin{cases} \{(-2\sqrt{y}, 3y),\ (-2\sqrt{y}, -5y)\}, & y \ge 0, \\ \varnothing, & y < 0. \end{cases} \tag{3.1.1}$$

For the unit proximal parameter, write $J = (I+F)^{-1}$. Then:

(a) The graph is closed, F is two-valued for $y > 0$, and

$$S = F^{-1}(0) = \mathbb{R} \times \{0\}, \qquad J(p,q) = \left(p + \sqrt{|q|}, \frac{|q|}{4}\right) \quad ((p,q) \in \mathbb{R}^2). \tag{3.1.2}$$

In particular, the complete proximal mapping is globally defined and single-valued.

(b) On every input set of diameter at most $D < +\infty$,

$$\|Jz - Jw\|^2 + \|(z - Jz) - (w - Jw)\|^2 \le \frac{1}{2}\left[\|z - w\|^2 + L_D^2\|z - w\|\right], \tag{3.1.3}$$

where

$$L_D = \left((2 + \sqrt{D})^2 + \frac{9}{4}D\right)^{1/2}. \tag{3.1.4}$$

Thus the corresponding complete graph block satisfies $\mathrm{RL}(1/2, L_D)$.

(c) At every $\bar{z} = (\bar{p}, 0) \in S$, the largest local RL power exponent is $1/2$. In particular, no finite local $\mathrm{RL}(1, L)$ *estimate holds on a full neighborhood of $\bar{z}$.*

(d) For $y \ge 0$,

$$r_F(\xi, y) := d(0, F(\xi, y)) = \sqrt{4y + 9y^2}, \qquad d((\xi, y), S) = y \le \frac{1}{4} r_F(\xi, y)^2. \tag{3.1.5}$$

The coefficient $1/4$ and the residual power 2 are asymptotically sharp. The exact gauge is

$$\psi_{\mathrm{ex}}(t) = \frac{\sqrt{16 + 36t^2} - 4}{18}. \tag{3.1.6}$$

Proof. The proximal equations for the two branches are

$$\begin{array}{ll} (+): & (p, q) = (\xi - 2\sqrt{y}, 4y), \quad q \ge 0, \\ (-): & (p, q) = (\xi - 2\sqrt{y}, -4y), \quad q \le 0. \end{array} \tag{3.1.7}$$

They cover the upper and lower input half-planes. At their intersection $q = 0$, both equations give $y = 0$, $\xi = p$, and the same graph value. Solving them gives the formula for J. Closedness follows from the two continuous branch parametrizations, and the zero set is immediate.

For the all-pairs estimate, let $z = (p, q)$, $w = (p', q')$, and put

$$a = p - p', \qquad b = q - q', \qquad h = \sqrt{|q|} - \sqrt{|q'|}, \qquad v = \frac{|q| - |q'|}{4}, \qquad \delta = \|z - w\|.$$

The proximal-output difference and the residual difference are $Jz - Jw = (a + h, v)$ and $(z - Jz) - (w - Jw) = (-h, b - v)$. Expanding their squared norms yields

$$\begin{aligned} E(z, w) &:= \|Jz - Jw\|^2 + \|(z - Jz) - (w - Jw)\|^2 \\ &= \frac{1}{2}\left[\delta^2 + (a + 2h)^2 + (2v - b)^2\right]. \end{aligned} \tag{3.1.8}$$

Since

$$|h| \le \sqrt{|q - q'|} \le \sqrt{\delta}, \qquad |2v - b| \le \frac{3}{2}|q - q'| \le \frac{3}{2}\delta,$$

we obtain

$$E(z, w) \le \frac{1}{2}\left[\delta^2 + (\delta + 2\sqrt{\delta})^2 + \frac{9}{4}\delta^2\right] \le \frac{1}{2}\left[\delta^2 + L_D^2\delta\right] \quad (\delta \le D). \tag{3.1.9}$$

The estimate includes pairs from opposite input half-planes.

To prove optimality, take $z_t = (\bar{p}, t)$ and $w_t = (\bar{p}, -4t)$, with $t > 0$. Direct substitution gives

$$\|z_t - w_t\|^2 = 25t^2, \qquad E(z_t, w_t) = 2t + \frac{269}{8}t^2. \tag{3.1.10}$$

For an RL estimate with exponent $\gamma > 1/2$, its right-hand side would be $O(t^2 + t^{2\gamma}) = o(t)$, contradicting this identity. Together with part (b), this proves (c).

The squared norms of the two values of $F(\xi, y)$ are $4y + 9y^2$ and $4y + 25y^2$. Their minimum proves the residual formula and the quadratic error bound. Solving $t^2 = 4y + 9y^2$ gives the exact gauge. Moreover,

$$\frac{y}{r_F(\xi, y)^2} = \frac{1}{4 + 9y} \longrightarrow \frac{1}{4} \qquad (y \downarrow 0).$$

If $y \le C r_F(\xi, y)^\theta$ held with $\theta > 2$, then $y / r_F(\xi, y)^\theta \asymp y^{1-\theta/2}$ would diverge. Thus both the coefficient and the power in part (d) are sharp in the stated sense. □

3.2 Application of the convergence theorem

Remark 3.2 (How the two certificates meet). Fix $\bar{z} = (\bar{p}, 0) \in S$. A ball $B_{D/2}(\bar{z})$ has input diameter at most D, and projection onto S preserves that ball. For z in this ball, write $d = d(z, S)$, $s = \|z - Jz\|$, and $d_+ = d(Jz, S)$. The RL step estimate and the independent error bound give

$$s \le \frac{1}{2}(d + L_D\sqrt{d}), \qquad d_+ \le \frac{1}{4}s^2.$$

Consequently,

$$d_+ \le \frac{(\sqrt{d} + L_D)^2}{16}\, d \le \kappa_D d, \qquad \kappa_D := \frac{(\sqrt{D} + L_D)^2}{16} \longrightarrow \frac{1}{4} \quad (D \downarrow 0). \tag{3.2.1}$$

For example, $D = 10^{-2}$ gives $\kappa_D < 0.305$. Choose the initial neighborhood to satisfy the localization budget in Theorem 2.6. That theorem then gives distance contraction, finite length, and convergence to one point of S. The proximal mapping is globally defined, so local existence requires no further argument in this example. The limiting certificate factor $1/4$ agrees with the exact distance factor calculated below.

3.3 Exact orbit and the two convergence scales

Proposition 3.3 (Normal contraction and summable tangential motion)**.** *For $z^0 = (p_0, q_0) \in \mathbb{R}^2$, set $r_0 = |q_0|$. The PPA orbit $z^{k+1} = Jz^k$ satisfies*

$$p_k = p_0 + 2\sqrt{r_0}(1 - 2^{-k}), \qquad |q_k| = 4^{-k} r_0, \tag{3.3.1}$$

where $q_k = 4^{-k} r_0$ for $k \ge 1$. Hence

$$d(z^{k+1}, S) = \frac{1}{4} d(z^k, S), \qquad z^k \longrightarrow z^\infty = (p_0 + 2\sqrt{r_0}, 0). \tag{3.3.2}$$

The total length obeys

$$\sum_{k=0}^{\infty} \|z^{k+1} - z^k\| \le \begin{cases} 2\sqrt{r_0} + r_0, & q_0 \ge 0, \\ 2\sqrt{r_0} + \dfrac{3}{2} r_0, & q_0 < 0. \end{cases} \tag{3.3.3}$$

For $k \ge 1$,

$$\|z^k - z^\infty\| = \left(4 r_0\, 4^{-k} + r_0^2\, 16^{-k}\right)^{1/2}. \tag{3.3.4}$$

Thus, if $r_0 > 0$, the asymptotic ratio of successive point errors is $1/2$, whereas the distance-to-set ratio is $1/4$. If $r_0 = 0$, the orbit is stationary.

Proof. The formula for J gives

$$q_{k+1} = \frac{|q_k|}{4}, \qquad p_{k+1} = p_k + \sqrt{|q_k|}.$$

Summing the geometric sequences proves the orbit and limit formulas. The horizontal variations sum to $2\sqrt{r_0}$. When $q_0 \geq 0$, the vertical variations sum to r_0. When $q_0 < 0$, the first vertical variation is $5r_0/4$, and those remaining sum to $r_0/4$. Bounding each step by the sum of its coordinate variations proves the length estimate. Subtraction from the limit gives the exact point error and its asymptotic ratio. □

The normal coordinate measures distance to the line of solutions, while the horizontal coordinate records motion along that line. A proximal step therefore has a tangential component of order $\sqrt{d(z^k, S)}$, larger than its current normal error. This component remains summable because the normal error decreases geometrically. The finite-length conclusion of the main theorem controls precisely this distinction between reaching the solution set and converging to one particular solution.

3.4 The multivalued difficulty and comparison with quadratic conditions

The two branches of F cover different proximal input half-planes. A complete verification must establish coverage, consistency where those half-planes meet, estimates for points from different branches, and an error bound for the minimum residual over both values. All four requirements are checked above. The square-root estimate accommodates the loss of a finite linear modulus at the common boundary. The same loss is already visible within either branch when approaching that boundary; joining the branches does not by itself create the fractional exponent.

Although F is multivalued, its proximal mapping is single-valued, so this example tests a multivalued graph with non-Lipschitz proximal geometry rather than random selection among proximal outputs.

Remark 3.4 (Exact scope of the separation). The divergence in (3.1.10) rules out every finite exponent-one RL estimate on the same full neighborhood. Proposition 4.2 gives the precise consequences for quadratic J-side, hypomonotone, almost-averaged, and fixed-anchor residual-quadratic certificates. The comparison is made certificate by certificate; it is not a claim that these broader theories cannot treat set-valued operators.

The additional scope exhibited here is therefore specific: nonlinear RL admits a complete multivalued graph whose proximal output and residual differences have square-root, rather than linear, control by the input difference. The independent quadratic error bound converts that control into contraction of the distance to the solution set. The explicit orbit also proves convergence directly and fixes the exact rates, providing a check on the general theorem.

4 Relation to existing theory

The comparison has two parts. At exponent one, RL recovers familiar quadratic operator inequalities and their PPA distance estimates. At an exponent below one, the square-root example separates the present assumptions from specific quadratic certificates on the same neighborhood. All of the operator frameworks compared below allow set-valued operators.

4.1 The quadratic endpoint

Proposition 4.1 (Identification of the exponent-one condition)**.** *Fix a graph block and $\lambda > 0$. For each tested pair of its graph points, write $a = u - v$ and $b = u^* - v^*$. The following statements hold on the same pair range as the RL condition.*

1. *The condition* $\mathrm{RL}(1, L)$ *is equivalent to*

$$\lambda\langle a, b\rangle \geq \frac{1 - L^2}{2(1 + L^2)}\big(\|a\|^2 + \lambda^2\|b\|^2\big). \tag{39}$$

Consequently $\mathrm{RL}(1, 1)$ *is precisely monotonicity on the block. In the notation of (μ, ν)-semimonotonicity,*

$$\mu = \frac{1 - L^2}{2\lambda(1 + L^2)}, \qquad \nu = \lambda^2\mu. \tag{40}$$

Thus $\mathrm{RL}(1, L)$ *is the balanced parameter slice of that inequality.*

2. *The scaled submonotonicity condition*

$$\lambda\langle a, b\rangle \geq -\tau\|a + \lambda b\|^2, \qquad \tau \geq 0,$$

is equivalent to $\mathrm{RL}(1, L)$ *with*

$$L^2 = 1 + 4\tau. \tag{41}$$

3. *If the block is h-hypomonotone, namely*

$$\langle a, b\rangle \geq -h\|a\|^2, \qquad h \geq 0,$$

and $\lambda h < 1$, then it satisfies $\mathrm{RL}(1, L_h)$ *with*

$$L_h = \frac{1 + \lambda h}{1 - \lambda h}. \tag{42}$$

Proof. At $\gamma = 1$, the RL inner-product inequality is

$$4\lambda\langle a, b\rangle \geq (1 - L^2)\|a + \lambda b\|^2.$$

Expanding the squared norm and collecting the inner-product terms proves (39). Setting $L = 1$ proves the monotone endpoint; dividing by λ gives (40). Substituting $1 - L^2 = -4\tau$ proves the second assertion.

For the last assertion, set $c = \lambda b$ and $\eta = \lambda h < 1$. Hypomonotonicity and Cauchy–Schwarz give

$$(1 - \eta)\|a\|^2 \leq \langle a, a + c\rangle \leq \|a\|\,\|a + c\|,$$

so $\|a\| \leq \|a + c\|/(1 - \eta)$. Therefore

$$\begin{aligned}\|a\|^2 + \|c\|^2 &= \|a + c\|^2 - 2\langle a, c\rangle \\ &\leq \left(1 + \frac{2\eta}{(1 - \eta)^2}\right)\|a + c\|^2 = \frac{1 + L_h^2}{2}\|a + c\|^2.\end{aligned}$$

This is the all-pairs J-side inequality with exponent one, under the identification $x = u + \lambda u^*$, $y = v + \lambda v^*$. □

The semimonotonicity parameters in (40) use the convention of [2]; (41) gives the scaled Luke–Tam condition [7]. Our local range condition (A1) uses the same coverage mechanism as [7, Assumption 2(d)(ii)] as its well-posedness component. The extension occurs in (A2): the exponent-one quadratic graph inequality is replaced by the nonlinear RL order, while (A3)–(A4) match that order with the output error bound.

4.2 Error bounds and the linear convergence factor

The regularity assumption concerns the original operator at the proximal output:

$$\operatorname{dist}(y, S) \le \psi\big(\operatorname{dist}(0, F(y))\big).$$

It can be verified independently of RL. In particular, an available metric q-subregularity estimate takes the form

$$\operatorname{dist}(y, S) \le K \operatorname{dist}(0, F(y))^q. \tag{43}$$

The orders $q > 0$, including $q > 1$, belong to the established subregularity theory [6, 9]. Substituting $\psi(t) = Kt^q$ into the direct recursion gives the coefficient $K(L/(2\lambda))^q$ and the leading power $d^{\gamma q}$ for $0 < \gamma < 1$, $L > 0$. This explains exactly which independently obtained error bound can be used: $\gamma q > 1$ gives local compatibility, while $\gamma q = 1$ requires $K(L/(2\lambda))^q < 1$.

General gauges also enter the abstract framework of [5, Theorems 4.3–4.4]. That framework concerns subgradient sequences with an objective descent condition, a surrogate for successive change, a relative subgradient-error bound, a continuity condition, and an accumulation point. Its regularity is written as a gauge applied to distance and is combined with the surrogate and relative-error moduli to obtain the rate map. Our theorem does not assume an underlying objective, descent, or a pre-existing accumulation point. Instead, the RL inequality supplies the step estimate, the proximal inclusion supplies the output residual, and an initial-radius budget proves localization and the existence of a limit. Thus the two frameworks overlap for suitable subgradient PPA specializations, but neither set of assumptions follows from the other at the present level of generality. The gauge assumption used here is an imported regularity condition; the square-root example verifies it by direct residual calculation.

Corollary 2.12 gives the coupled factor

$$\kappa_E = \sqrt{\frac{1 + L^2}{2}}\, \frac{\rho}{\sqrt{\rho^2 + \lambda^2}}.$$

At $L = 1$, this is $\rho/\sqrt{\rho^2 + \lambda^2}$, the distance factor in Leventhal's Theorem 3.1 [4]. Under the Luke–Tam normalization $L^2 = 1 + 4\tau$,

$$\kappa^2_{\text{coupled}} = (1 + 2\tau)\frac{\rho^2}{\rho^2 + \lambda^2}, \qquad \kappa_{\text{coupled}} < 1 \quad \Longleftrightarrow \quad 2\tau\rho^2 < \lambda^2. \tag{44}$$

For the common scaled submonotonicity inequality, the estimate displayed in [7] is

$$\kappa^2_{\text{LT}} = 1 + 2\tau - \left(\frac{\lambda}{\lambda + \rho}\right)^2, \qquad \tau(1 + \rho/\lambda)^2 < \frac{1}{2}.$$

Keeping the error bound at the output gives (44), with a smaller factor and a weaker scalar restriction. This comparison is confined to the common transition inequalities; coverage and localization must also hold in either local convergence theorem.

4.3 What the multivalued example separates

Almost-averaged mapping theory explicitly allows set-valued iteration maps [8]. Generalized-monotonicity PPA theory also allows set-valued operators [7, 2]. The distinction in the following proposition is the order of the estimate near the branch junction.

Proposition 4.2 (Failure of specified quadratic certificates)**.** *Let F be the square-root operator in Theorem 3.1, let $J = J_F$, and fix $\bar{z} = (\bar{p}, 0) \in S$. On every full neighborhood of $\bar{z}$:*

1. *there is no finite C such that*

$$\|Jz - Jw\|^2 + \|(z - Jz) - (w - Jw)\|^2 \le C\|z - w\|^2 \tag{45}$$

   for all inputs z, w in that neighborhood;

2. *there is no finite C such that $\|Jz - \bar{z}\| \le C\|z - \bar{z}\|$ for all nearby inputs;*

3. *there is no finite local hypomonotonicity modulus for the graph of F near $(\bar{z}, 0)$;*

4. *there is no fixed finite matrix V such that*

$$\langle u - \bar{z}, u^* \rangle \ge \langle u^*, Vu^* \rangle \tag{46}$$

   for all graph points (u, u^*) sufficiently near $(\bar{z}, 0)$.

Proof. The resolvent is $J(p, q) = (p + \sqrt{|q|}, |q|/4)$. Take $z_t = (\bar{p}, t)$ and $w_t = (\bar{p}, -4t)$, $t > 0$. Direct calculation gives

$$\|z_t - w_t\|^2 = 25t^2, \qquad \|Jz_t - Jw_t\|^2 + \|(z_t - Jz_t) - (w_t - Jw_t)\|^2 = 2t + \frac{269}{8}t^2.$$

Their ratio diverges, proving the first assertion. For the second,

$$\frac{\|Jz_t - \bar{z}\|}{\|z_t - \bar{z}\|} = \frac{\sqrt{t + t^2/16}}{t} \longrightarrow +\infty.$$

To prove the third assertion, use the graph points $u_t = (\bar{p} + t, t)$ and $u_t^* = (-2\sqrt{t}, 3t)$, together with $(\bar{z}, 0)$. They satisfy

$$\frac{\langle u_t - \bar{z}, u_t^* \rangle}{\|u_t - \bar{z}\|^2} = -\frac{1}{\sqrt{t}} + \frac{3}{2} \longrightarrow -\infty.$$

Finally, replace u_t by $(\bar{p} + t^{1/4}, t)$, keeping the same u_t^*. Then

$$\frac{\langle u_t - \bar{z}, u_t^* \rangle}{\|u_t^*\|^2} = \frac{-2t^{3/4} + 3t^2}{4t + 9t^2} \longrightarrow -\infty.$$

For a fixed finite matrix V, the quotient $\langle u_t^*, Vu_t^* \rangle / \|u_t^*\|^2$ is bounded below by $-\|V\|$. This proves the last assertion. □

The four assertions exclude the displayed quadratic certificates on the same full neighborhood. They do not compare the entire semimonotone, weak-Minty, almost-averaged, or partial-regularity frameworks, which may use different relative domains or degenerate tests. In contrast, the Hölder J-side estimate and the quadratic output error bound of Theorem 3.1 satisfy the RLEB compatibility condition.

5 Verification from graph derivatives

The main theorem can be applied by checking its assumptions directly, as in the square-root example. In more complicated problems, an explicit formula for $J_{\lambda F}$ may be unavailable. A pair of smooth parametrizations of the operator graph can then provide a useful alternative. We describe this verification route here; its full statement and proof are in Appendix A.

Suppose the relevant graph consists of two parametrized branches

$$g_\sigma(a,t) = (u_\sigma(a,t), v_\sigma(a,t)), \qquad \sigma \in \{+,-\}, \quad t \geq 0,$$

which agree at $t = 0$. Their proximal inputs are

$$X_\sigma(a,t) = u_\sigma(a,t) + \lambda v_\sigma(a,t).$$

The parameter a runs along the seam, while t measures displacement from it on each branch. The method first solves the tangential components of $X_\sigma(a,t) = x$ for a. The remaining scalar input coordinate is denoted by $n_\sigma(z,t)$, where z is the fixed tangential input. The key hypothesis requires the two branches to leave their common input seam in opposite directions:

$$\partial_t n_+(z,t) \geq p c_+ t^{p-1}, \qquad -\partial_t n_-(z,t) \geq p c_- t^{p-1}, \qquad p \geq 1, \quad c_+, c_- > 0.$$

These inequalities are checked uniformly over the parameter domain. The derivative remaining after tangential elimination is a Schur complement; the two signs distinguish the sides of the seam.

Under the tangential invertibility and derivative bounds of Theorem A.1, together with its complete-fiber condition, this calculation gives an open input neighborhood U with the following properties. Every $x \in U$ has exactly one localized proximal output Jx, and comparisons across the two branches satisfy

$$\|Jx - Jx'\|^2 + \|(x - Jx) - (x' - Jx')\|^2 \leq \frac{1}{2}\left(\|x - x'\|^2 + L_D^2 \|x - x'\|^{2/p}\right) \tag{47}$$

whenever x, x' belong to a covered input block of diameter at most D. Thus the produced RL exponent is $\gamma = 1/p$. The constant is explicit: for $p > 1$, put $p' = p/(p-1)$; then

$$L_D = K_z D^{1-1/p} + (1 + B^2)^{1/(2p)} \left[(K_+ c_+^{-1/p})^{p'} + (K_- c_-^{-1/p})^{p'}\right]^{1/p'}.$$

Here B, K_z, K_+, K_- are the uniform derivative bounds specified in (S3)–(S5). Theorem A.1 also gives the size of U, and Proposition A.3 improves the constant when the two branch derivatives match to higher order.

Completeness of the represented graph is part of the verification: every graph point with an input in U must be accounted for. Opposite signed growth then rules out two distinct outputs at the same input, including outputs from different branches. The theorem concerns the localized resolvent unless completeness is established for the unrestricted graph.

This procedure supplies coverage, uniqueness, and the J-side geometry. The residual error bound is obtained separately and combined with these data through the main convergence theorem. The proof uses tangential inversion and one-dimensional growth; its detailed hypotheses and constants are retained in the appendix for applications in which branch data are easier to obtain than the resolvent itself.

6 Conclusion

The RL condition extends the inner-product inequality of monotonicity by a nonlinear term. Its J-side form gives an energy estimate and the step bound $s \le (d + Ld^\gamma)/2$. The proximal inclusion then bounds the residual at the next iterate by s/λ, and generalized metric subregularity gives the distance recursion $d^+ \le \psi((d + Ld^\gamma)/(2\lambda))$. Theorem 2.6 turns local contraction of this scalar map into finite length, localization, and convergence of the whole orbit to a single solution. At the critical residual-growth order $a = \gamma < 1$, the condition is $\lambda m > L/2$. The complementary energy estimate also retains the known monotone-PPA rate at the linear endpoint.

The two-valued operator in Theorem 3.1 satisfies the nonlinear condition at a branch seam. Proposition 3.3 gives its exact proximal orbit. Its best RL exponent is $1/2$, and its quadratic residual error bound has the matching order. The resulting proximal orbit approaches the nonisolated zero set with exact factor $1/4$, while every nonstationary orbit has asymptotic pointwise factor $1/2$. The explicit trajectory identifies the source of these two scales: normal contraction is accompanied by a larger, summable tangential motion. The general RLEB distance estimate approaches the exact factor $1/4$ as the verification neighborhood shrinks.

Graph-level RL requires comparisons across every pair of relevant branches. The square-root example satisfies these comparisons at Hölder order but not at exponent one. Corollary 2.7 retains the same convergence estimates for genuinely set-valued proximal relations when the one-step inequalities hold uniformly over all selections. The standard range condition (A1) supplies local PPA coverage, while (A3)–(A4) state the precise regularity strength needed to convert nonlinear graph geometry into convergence.

The signed-Schur certificate is one additional way to verify the geometric hypotheses from graph parametrizations. For complete two-sided seams with one degenerate normal direction, it supplies a covered input neighborhood, cross-branch uniqueness, and explicit RL parameters. It can be used when graph data are available without a closed formula for the proximal mapping. It supplies a local module for constructing finite branch atlases and for studying higher-codimension junctions.

A Signed-Schur verification on two complete graph branches

This appendix supplies the derivative conditions and proof for the verification procedure in Section 5. Its input is a pair of graph parametrizations. Tangential elimination reduces coverage and uniqueness to a scalar growth test with opposite signs on the two branches. The resulting estimates provide the J-side inequality; the error bound is verified separately.

Fix $\lambda > 0$. For graph points $g = (u, u^*)$, use the algebraic coordinates

$$\mathsf{X}g = u + \lambda u^*, \qquad \mathsf{Y}g = u - \lambda u^*.$$

Choose orthogonal input coordinates $\mathbb{R}^n = \mathbb{R}^{n-1} \times \mathbb{R}$, with projections P and N. Norms below are Euclidean operator norms.

Theorem A.1 (Signed Schur-growth certificate)**.** *Let $n \ge 2$ and $R, T, \mu > 0$. Fix nonnegative constants H, B, K_z, K_+, K_-. Suppose that two C^1 graph parametrizations*

$$g_\sigma(a,t) = (u_\sigma(a,t), v_\sigma(a,t)), \qquad (a,t) \in \overline{B}_R^{n-1} \times [0,T], \quad \sigma \in \{+1,-1\},$$

are defined on a neighborhood of their displayed parameter domains and take values in $\operatorname{gph} F$*. They agree on the seam:*

$$g_+(a,0) = g_-(a,0) = g_0(a). \tag{S1}$$

Write $X_\sigma = \mathsf{X} g_\sigma$, $Y_\sigma = \mathsf{Y} g_\sigma$, and assume $PX_\sigma(0,0) = 0$. At every displayed parameter point, form

$$A_\sigma = PD_a X_\sigma, \qquad b_\sigma = P\partial_t X_\sigma.$$

Assume the following bounds hold uniformly on both complete parameter domains:

$$\operatorname{sym} A_\sigma \succeq \mu I, \qquad \|b_\sigma\| \le H, \tag{S2}$$

$$\|N D_a X_\sigma A_\sigma^{-1}\| \le B, \tag{S3}$$

$$\|D_a Y_\sigma A_\sigma^{-1}\| \le K_z, \tag{S4}$$

$$\|\partial_t Y_\sigma - D_a Y_\sigma A_\sigma^{-1} b_\sigma\| \le K_\sigma. \tag{S5}$$

Let $h_+, h_- : [0,T] \to [0,\infty)$ be C^1, convex, strictly increasing, and zero at zero. The signed normal Schur derivatives are required to satisfy

$$\sigma\big(N\partial_t X_\sigma - N D_a X_\sigma A_\sigma^{-1} b_\sigma\big) \ge h_\sigma'(t). \tag{S6}$$

Choose $0 < r < \mu R - HT$. Then the following conclusions hold.

1. *For every $z \in B_r^{n-1}$ and $t \in [0,T]$, there is exactly one $a = \alpha_\sigma(z,t) \in B_R^{n-1}$ with $PX_\sigma(a,t) = z$. The seam function*

   $$\nu(z) = N X_\sigma(\alpha_\sigma(z,0),0)$$

   is independent of σ. The represented graph covers the open input neighborhood

   $$U = \{(z,s) : z \in B_r^{n-1}, \quad \nu(z) - h_-(T) < s < \nu(z) + h_+(T)\}. \tag{S7}$$

2. *On the represented graph with tangential inputs in B_r^{n-1}, X has no collisions: equal inputs imply equal graph points. At the seam, the two parameter representations describe the same graph point, not two different solutions.*

3. *Set $C_B = \sqrt{1+B^2}$, and define*

   $$V(v) = \max\{K_+ s + K_- t : 0 \le s,t \le T,\ h_+(s) + h_-(t) \le v\}, \qquad v \ge 0. \tag{S8}$$

   For every pair of represented graph points $g = (u,v)$ and $g' = (u',v')$ whose tangential inputs belong to B_r^{n-1},

   $$\begin{gathered} \|u-u'\|^2 + \lambda^2 \|v-v'\|^2 \le \frac{1}{2}\big[d^2 + \{K_z d + V(C_B d)\}^2\big], \\ d = \|\mathsf{X} g - \mathsf{X} g'\|, \end{gathered} \tag{S9}$$

   holds. The bound $\omega(d) = K_z d + V(C_B d)$ is nondecreasing and continuous at zero. In the power case below it gives explicit RL parameters. On the covered input neighborhood, the left-hand side is $\|Jx - Jx'\|^2 + \|(x - Jx) - (x' - Jx')\|^2$.

If the intended graph localization $\mathcal{G}$ satisfies the complete fiber identity

$$\mathcal{G} \cap \mathsf{X}^{-1}(U) = \left(\bigcup_{\sigma=\pm 1} g_\sigma(\overline{B}_R^{n-1} \times [0,T]) \right) \cap \mathsf{X}^{-1}(U), \tag{S10}$$

then all conclusions apply to that localization of F, including a single-valued localized resolvent on all of U. For an unrestricted resolvent conclusion, $\mathcal{G}$ must be the complete graph of F over these inputs.

Proof. Fix t and σ, and put $f(a) = PX_\sigma(a,t)$. Integration of (S2) along a line segment gives

$$\langle f(a) - f(a'), a - a' \rangle \geq \mu \|a - a'\|^2. \tag{S11}$$

Also $\|f(0)\| \leq HT$. For $\|a\| = R$, $\|z\| < r$,

$$\langle f(a) - z, a \rangle \geq \mu R^2 - (HT + r)R > 0.$$

The continuous map $f - z$ therefore has a zero in the ball. Indeed, if it did not, the self-map $a \mapsto -R(f(a) - z)/\|f(a) - z\|$ of the closed ball would have a Brouwer fixed point on the sphere, contradicting the displayed positive inner product. Uniqueness follows from (S11). Since A_σ is invertible, the parameterized implicit function theorem yields a C^1 map α_σ, with

$$D_z \alpha_\sigma = A_\sigma^{-1}, \qquad \partial_t \alpha_\sigma = -A_\sigma^{-1} b_\sigma. \tag{S12}$$

All matrices on the right are evaluated at the corrected parameter $(\alpha_\sigma(z,t), t)$. At the seam,

$$\alpha_+(z,0) = \alpha_-(z,0).$$

Define, solely for this proof,

$$n_\sigma(z,t) = NX_\sigma(\alpha_\sigma(z,t),t), \qquad y_\sigma(z,t) = Y_\sigma(\alpha_\sigma(z,t),t).$$

The chain rule, (S12), and the hypotheses give

$$\|D_z n_\sigma\| \leq B, \quad \sigma \partial_t n_\sigma \geq h_\sigma'(t), \quad \|D_z y_\sigma\| \leq K_z, \quad \|\partial_t y_\sigma\| \leq K_\sigma. \tag{S13}$$

Both normal coordinates start at $\nu(z)$. Integration shows that $n_+(z,\cdot)$ is strictly increasing and reaches at least $\nu(z) + h_+(T)$; $n_-(z,\cdot)$ is strictly decreasing and reaches at most $\nu(z) - h_-(T)$. The intermediate value theorem proves (S7). Their ranges meet only at the seam, and each is one-to-one. Consequently equal full Minty inputs imply the same branch parameter, except for the duplicate seam representation, where (S1) gives the same graph point.

For convex h with $h(0) = 0$,

$$h(t) - h(s) \geq h(t - s), \qquad 0 \leq s \leq t \leq T. \tag{S14}$$

Consider two points on the same branch with reduced parameters (z,t) and (w,s). Equations (S13)–(S14) give

$$h_\sigma(|t - s|) \leq |n_\sigma(z,t) - n_\sigma(w,s)| + B\|z - w\| \leq C_B d.$$

The difference of the Y-coordinates is at most $K_z\|z - w\| + K_\sigma|t - s|$, which is bounded by $K_z d + V(C_B d)$, since the pair $(|t - s|, 0)$, or $(0, |t - s|)$, is feasible in (S8).

For points on opposite branches, use the common seam at the same tangential coordinate to obtain

$$h_+(t) + h_-(s) \leq n_+(z,t) - n_-(z,s) \leq |n_+(z,t) - n_-(w,s)| + B\|z - w\| \leq C_B d.$$

Compare their Y-coordinates through $y_0(z)$ and $y_0(w)$:

$$\|y_+(z,t) - y_-(w,s)\| \leq K_+ t + K_z\|z - w\| + K_- s \leq K_z d + V(C_B d).$$

Thus $\|\mathsf{Y}g - \mathsf{Y}g'\| \leq K_z d + V(C_B d)$ for every ordered branch pair, including seam endpoints. The identity

$$2\big(\|u - u'\|^2 + \lambda^2\|v - v'\|^2\big) = \|\mathsf{X}g - \mathsf{X}g'\|^2 + \|\mathsf{Y}g - \mathsf{Y}g'\|^2$$

gives (S9). The maximization defining V is over a nonempty compact set, and strict increase of both h_σ forces $V(v) \to 0$ as $v \downarrow 0$. The complete-fiber conclusion is then immediate. □

Corollary A.2 (Explicit power constants)**.** *In Theorem A.1, suppose $h_\sigma(t) = c_\sigma t^p$, with $c_\sigma > 0$, $p > 1$. Put $p' = p/(p-1)$ and*

$$C_p = \left[(K_+ c_+^{-1/p})^{p'} + (K_- c_-^{-1/p})^{p'} \right]^{1/p'}.$$

On every represented input block of diameter at most $D > 0$,

$$\boxed{\gamma = 1/p, \qquad L = K_z D^{1-1/p} + (1+B^2)^{1/(2p)} C_p.} \tag{S15}$$

For $p = 1$, replace C_p by $\max\{K_+/c_+, K_-/c_-\}$ and take $\gamma = 1$. If the branch powers differ, $h_\sigma(t) = c_\sigma t^{p_\sigma}$, one may take $p = \max\{p_+, p_-\}$ and replace c_σ by $c_\sigma T^{p_\sigma - p}$ in the same formula.

Proof. Hölder's inequality in $\mathbb{R}^2$ gives $V(v) \le C_p v^{1/p}$, without requiring that the maximizing parameters remain below T. Insert this in (S9) and use $d \le D^{1-1/p} d^{1/p}$. The $p = 1$ formula is the elementary linear-programming bound. Finally, $t^{p_\sigma} \ge T^{p_\sigma - p} t^p$ for $0 \le t \le T$. □

Proposition A.3 (A matched-jet improvement across the seam)**.** *Suppose the hypotheses of Theorem A.1 hold with $h_+(t) = h_-(t) = ct^p$, $c > 0$, $p \ge 1$. In the reduced coordinates constructed in its proof, assume the additional paired-branch estimate*

$$\|y_+(z,t) - y_-(z,t)\| \le E t^p \quad (z \in B_r^{n-1},\ 0 \le t \le T). \tag{S16}$$

Let $K = \max\{K_+, K_-\}$. In the J-side estimate (S9)*, the term $K_z d + V(C_B d)$ can be replaced by*

$$\omega(d) = \left(K_z + \frac{E C_B}{2c} \right) d + K \left(\frac{C_B}{c} \right)^{1/p} d^{1/p}. \tag{S17}$$

In particular, on an input block of diameter D, a valid constant is

$$L_D = \left(K_z + \frac{E C_B}{2c} \right) D^{1-1/p} + K (C_B/c)^{1/p}.$$

Condition (S16) *follows, for example, by integrating $\|\partial_t y_+(z,t) - \partial_t y_-(z,t)\| \le pEt^{p-1}$. Both reduced derivatives are the native Schur expressions in (S5), evaluated at their corresponding tangentially corrected parameters.*

Proof. Same-branch pairs already satisfy the stated bound. For opposite branches, suppose $t \ge s$, compare first on the branch with parameter t, and then use (S16) at s:

$$\|y_+(z,t) - y_-(w,s)\| \le K_z \|z - w\| + K|t - s| + E s^p.$$

If $s \ge t$, interchange the branches and use the seam mismatch at t. The opposite-branch estimate in the theorem gives $c(t^p + s^p) \le C_B d$, whence

$$|t - s| \le (C_B d/c)^{1/p}, \qquad \min\{t, s\}^p \le C_B d/(2c).$$

Substitution proves the claim. □

Proposition A.4 (When the produced exponent is genuine)**.** *Under the preceding corollary, fix a tangential coordinate z_0. Suppose that on at least one branch,*

$$|n_\sigma(z_0, t) - \nu(z_0)| \le C t^p, \qquad \|y_\sigma(z_0, t) - y_\sigma(z_0, 0)\| \ge ct$$

for $0 < t < t_0$, with $C, c > 0$. Then no exponent greater than $1/p$ is valid on a graph neighborhood containing this arc and its seam point. In particular, for $p > 1$ the localized resolvent is not calm there and hence is not locally Lipschitz.

Proof. Along this fixed tangential-input arc, an RL estimate of exponent θ would give $ct \le LC^\theta t^{p\theta}$. This is impossible as $t \downarrow 0$ when $p\theta > 1$. Moreover, $2(u-u_0) = (\mathsf{X}g - \mathsf{X}g_0) + (\mathsf{Y}g - \mathsf{Y}g_0)$ implies $2\|u-u_0\| \ge ct - Ct^p$, which excludes calmness of the resolvent for $p > 1$. □

Remark A.5 (Computable hypotheses and a finite atlas). The Schur quantities in (S2)–(S6) are evaluated directly on graph parametrizations. No normal inverse or resolvent formula is a hypothesis. If a graph branch is specified by an implicit active-manifold equation $G_\sigma(a,t,w) = 0$, a certified invertible $D_w G_\sigma$ produces its derivatives through $D_{(a,t)}w = -(D_w G_\sigma)^{-1} D_{(a,t)} G_\sigma$. Interval bounds on these matrices can therefore certify (S2)–(S6), provided branch existence and the complete active-set list have been established independently.

Several seam neighborhoods can be combined using a finite-chart chaining argument, but that additional step still requires compatible complete fibers on overlaps and a quantitative input-chain geometry. The present theorem supplies a nontrivial *local* derivative certificate at a corank-one two-sided seam. It is not a theorem for arbitrary higher-codimension Whitney stratifications or arbitrarily many branches meeting on the same side.

Proposition A.6 (Native verification of the square-root operator)**.** *Let* $\lambda = 1$ *and*

$$F(\xi,y) = \{(-2\sqrt{y}, 3y), (-2\sqrt{y}, -5y)\} \quad (y \ge 0), \qquad F(\xi,y) = \varnothing \quad (y < 0).$$

Use the native, smooth graph parameters

$$g_+(a,t) = ((a,t^2),(-2t,3t^2)), \qquad g_-(a,t) = ((a,t^2),(-2t,-5t^2)), \quad t \ge 0.$$

On $|a| \le R$, $0 \le t \le T$, *the Schur data can be chosen as*

$$\mu = 1, \quad H = 2, \quad B = 0, \quad K_z = 1, \quad h_+(t) = h_-(t) = 4t^2,$$

$$K_+ = 4\sqrt{1+T^2}, \qquad K_- = 4\sqrt{1+9T^2}.$$

For every $0 < r < R - 2T$, *the complete graph over*

$$U = (-r, r) \times (-4T^2, 4T^2)$$

is covered uniquely in Minty inputs. On an input subblock of diameter D, *the basic theorem yields*

$$\gamma = \tfrac{1}{2}, \qquad L_D = \sqrt{D} + \sqrt{8 + 40T^2}.$$

The matched-jet version improves the asymptotic leading constant:

$$\gamma = \tfrac{1}{2}, \qquad L_D^{\mathrm{jet}} = 2\sqrt{D} + 2\sqrt{1+9T^2}. \tag{S18}$$

In particular, along shrinking collars $D, T \to 0$, $L_D^{\mathrm{jet}} \to 2$, *which is the optimal asymptotic RL constant. These bounds are produced from graph derivatives without assuming or solving a full inverse formula.*

Proof. Direct calculation on the native graph gives

$$X_\pm(a,t) = (a - 2t,\ \pm 4t^2), \qquad Y_+(a,t) = (a+2t, -2t^2), \quad Y_-(a,t) = (a+2t, 6t^2).$$

Thus $A_\pm = 1$, $b_\pm = -2$, and the signed normal Schur derivatives are $8t$. The output Schur derivatives are $(4, -4t)$ and $(4, 12t)$, giving the asserted constants. The common tangential

correction is $a = z + 2t$. Consequently the two corrected output jets differ by $(0, -8t^2)$, so that (S16) holds with $E = 8$. Formula (S18) follows from Proposition A.3.

For completeness of the represented fibers, any full graph point with Minty input in U has $t = \sqrt{y} < T$ and $|a| = |z + 2t| < r + 2T < R$, and hence belongs to one of the displayed graph parametrizations. Conversely, the two monotone normal ranges cover $(-4T^2, 4T^2)$. Finally, along $z = 0$, on the plus branch, the input difference from the seam is $4t^2$ and the output difference is $\sqrt{16t^2 + 4t^4}$. Their RL ratio at exponent $1/2$ tends to 2, and every larger exponent fails. □

Proposition A.7 (Branchwise regularity does not replace orientation)**.** *Even smooth, semi-algebraic graph branches with a common seam, individually one-to-one Minty maps, and individual optimal Hölder inverses need not satisfy any all-pairs RL inequality after union.*

Proof. In one input dimension, prescribe the two Minty-coordinate branches

$$X_+(t) = X_-(t) = t^2, \qquad Y_+(t) = t, \quad Y_-(t) = -t, \qquad t \geq 0.$$

They define operator graph branches through $u_\sigma = (X_\sigma + Y_\sigma)/2$, $v_\sigma = (X_\sigma - Y_\sigma)/(2\lambda)$. Each branch is smooth and its Minty map is one-to-one, with $|Y_\sigma(t) - Y_\sigma(s)| \leq |X_\sigma(t) - X_\sigma(s)|^{1/2}$. Both branches agree at zero. Their union is a smooth parabola in (X, Y)-coordinates; splitting it into its two open arcs and the origin is a Whitney stratification. Nevertheless, at any $t > 0$, the two inputs agree, while the primal outputs differ by t. Thus every all-pairs RL inequality fails. The missing hypothesis is precisely opposite signed normal growth, not individual branch smoothness or stratifiability. □

Remark A.8 (Uniformity in the tangential parameter is essential). The smooth input map $X(a, t) = (a, t^3 - at)$ has the cubic normal law $t \mapsto t^3$ at $a = 0$, but for each small $a > 0$ its normal coordinate is zero at both $t = 0$ and $t = \sqrt{a}$. Taking $Y(a, t) = (a, t)$ again defines a legitimate operator graph with a Minty collision. A Taylor expansion at a single seam point cannot replace the uniform Schur lower bound (S6).

Relation to inverse-function machinery. The proof combines strong-monotonicity inversion in tangential coordinates, a parameterized implicit-function calculation, and one-dimensional monotone growth. Its RLEB-specific output is the joint quantitative certificate for a complete two-sided graph: a covered input collar, cross-branch collision exclusion, and the constants in (S15). Whitney stratification alone is insufficient, as Proposition A.7 shows, and the residual error bound must still be verified separately.

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